

National Institute of Technology Rourkela

Department of Computer Science and Engineering

Mid-Term Examination (Feb 2024)

Deep Learning (CS6510), B.Tech. 8th Semester

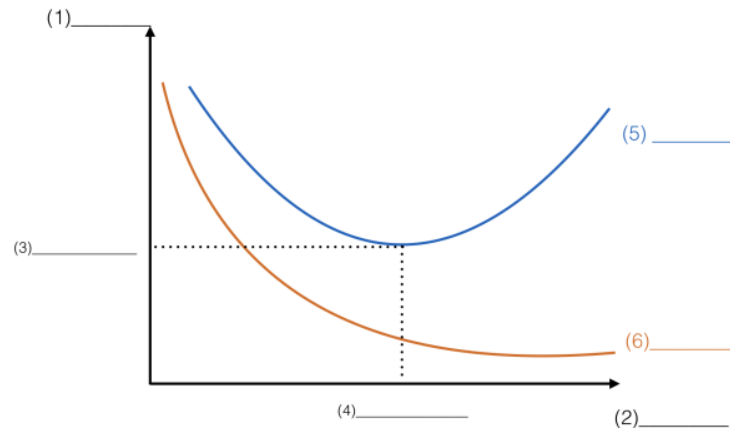


Time: 120 Minutes

Max. Marks: 30

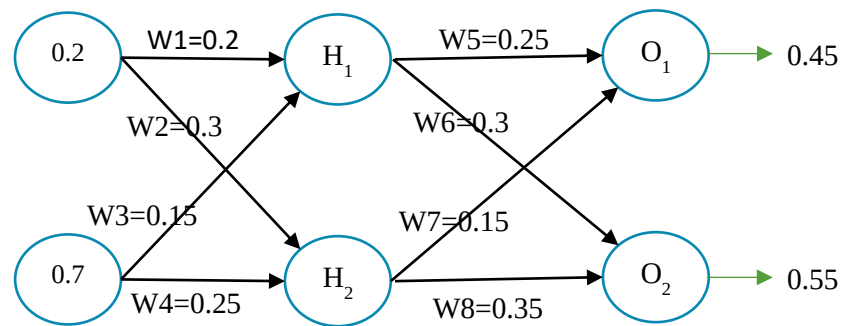
Instruction: 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. There are 09 questions printed on the paper. Attempt all the questions.

- Q1. A 1-hidden layer neural network with k neurons, n inputs and m outputs with bias at each neuron will have how many total parameters. **01**
- Q2. Consider a fully connected network with n inputs $x_1, x_2, \dots, x_n$. Suppose there is a single hidden layers with k neurons having f as activation function and b_j as the bias of j th hidden neuron. Further, the output layer contains a single neuron with g as activation function and B as the bias. Assume that the weights between inputs to hidden layer is denoted by w and weights between hidden to output layer is denoted as v . Write down the output y of the network as a function of x , weights and biases. **02**
- Q3. You come across a nonlinear function that passes 4 if its input is nonnegative, else evaluates to -2, i.e. $f(x) = \begin{cases} 4 & \text{if } x \geq 0 \\ -2 & \text{Otherwise} \end{cases}$ **02**
A friend recommends you to use this non-linearity in your deep neural network with the Adam optimizer. Would you follow his advice? Why or why not?
- Q4. You want to use the figure below to explain the concept of early stopping to a friend. Fill-in the blanks. (1) and (2) describe the axes. (3) and (4) describe values on the vertical and horizontal axis. (5) and (6) describe the curves. Be precise. **03**



- Q5. What is the difference between gradient descent, stochastic gradient descent and mini-batch stochastic gradient descent? Given a dataset having 10020 input instances. How many times the weights will be updated in one epoch if Gradient Descent, Stochastic Gradient Descent and mini-batch stochastic gradient descent with 128 batch size are separately used? **05**

- Q6. If the sets P and N are finite and linearly separable, prove that the perceptron learning algorithm updates the weight vector w_t a finite number of times such that the two sets are separated. **04**
- Q7. Prove that moving in the opposite direction to the function Gradient minimizes the function. Explain the Backpropagation algorithm using gradient descent update rule. **04**
- Q8. What problem occurs if the learning rate is too high or too low? Why adaptive learning rate with respect to cumulative sum of gradients are used in Adagrad optimizer? Explain how the problem of Adagrad is overcome in RMSProp by mentioning the weight update equations? **05**
- Q9. Consider the following neural network architecture with sigmoid activation function at every neuron and given weights as initial weights, inputs and outputs. Compute the value of W_4 , W_7 and W_8 after 1 epoch of backpropagation algorithm using stochastic gradient descent weight updating rule with learning rate is 0.1. **04**



*****End of Question Paper*****

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

MID - SEM EXAMINATION, 2025

SESSION: 2024 – 2025 (Spring)

M.Tech. 2nd Semester and Ph.D

Subject code: CS6592

No. of pages: 2

Subject Name: Applied Deep Learning

Full Marks: 30

Dept. Code: CS

Duration: 2 Hrs

Answer ALL questions. Unnecessary and extra writing will attract negative marks.

Q. No.	Questions	Marks	CO	BL														
1. (a)	A CNN apply a 2×2 kernel (filter) on a 3×3 image with padding = 1. Compute the output. What will be the final result if we add a 2×2 max pooling operation to the previous result? Image <table><tr><td>4</td><td>5</td><td>3</td></tr><tr><td>9</td><td>8</td><td>6</td></tr><tr><td>6</td><td>2</td><td>1</td></tr></table> Kernel <table><tr><td>1</td><td>3</td></tr><tr><td>4</td><td>6</td></tr></table> (b) Explain Exploding Gradient and Vanishing Gradient problem. How can these be solved? <tr><td>2. (a)</td><td> Show one training iteration of the given neural network where $x_1 = 1$, $x_2 = 1$, and desired output $y_{d,5} = 0$. Assume the learning rate is 0.1. All the bias (θ) is connected to a fixed input -1. The initial weights and threshold are as follows: $w_{13} = 0.5$, $w_{14} = 0.9$, $w_{23} = 0.4$, $w_{24} = 1.0$, $w_{35} = -1.2$, $w_{45} = 1.1$, $\theta_3 = 0.8$, $\theta_4 = -0.1$ and $\theta_5 = 0.3$. Using the ReLU activation function, what will be the updated weights and threshold after two iterations? The diagram shows a neural network with three layers: Input layer, Hidden layer, and Output layer. The input layer has two nodes, labeled 1 and 2, with inputs x_1 and x_2 respectively. The hidden layer has two nodes, labeled 3 and 4. The output layer has one node, labeled 5, with output y_5. Each node in the input, hidden, and output layers is connected to a bias node with value -1. The weights are labeled w_{13}, w_{14}, w_{23}, w_{24}, w_{35}, and w_{45}. The thresholds are labeled θ_3, θ_4, and θ_5. (b) What is overfitting and underfitting? Write a solution for the overfitting problem. <tr><td>3. (a)</td><td> Explain Adam optimizer algorithm using the function $f(w) = w^2$. Take the initial weight(w) is 2, learning rate 0.1, and exponential decay rates for the moments $\beta_1 = 0.9$ and $\beta_2 = 0.999$. What is the advantage of Momentum over SGD? (b) What is SoftMax activation? How is it different from the Sigmoid function? </td></tr></td></tr>	4	5	3	9	8	6	6	2	1	1	3	4	6	2. (a)	Show one training iteration of the given neural network where $x_1 = 1$, $x_2 = 1$, and desired output $y_{d,5} = 0$. Assume the learning rate is 0.1. All the bias (θ) is connected to a fixed input -1. The initial weights and threshold are as follows: $w_{13} = 0.5$, $w_{14} = 0.9$, $w_{23} = 0.4$, $w_{24} = 1.0$, $w_{35} = -1.2$, $w_{45} = 1.1$, $\theta_3 = 0.8$, $\theta_4 = -0.1$ and $\theta_5 = 0.3$. Using the ReLU activation function, what will be the updated weights and threshold after two iterations? The diagram shows a neural network with three layers: Input layer, Hidden layer, and Output layer. The input layer has two nodes, labeled 1 and 2, with inputs x_1 and x_2 respectively. The hidden layer has two nodes, labeled 3 and 4. The output layer has one node, labeled 5, with output y_5. Each node in the input, hidden, and output layers is connected to a bias node with value -1. The weights are labeled w_{13}, w_{14}, w_{23}, w_{24}, w_{35}, and w_{45}. The thresholds are labeled θ_3, θ_4, and θ_5. (b) What is overfitting and underfitting? Write a solution for the overfitting problem. <tr><td>3. (a)</td><td> Explain Adam optimizer algorithm using the function $f(w) = w^2$. Take the initial weight(w) is 2, learning rate 0.1, and exponential decay rates for the moments $\beta_1 = 0.9$ and $\beta_2 = 0.999$. What is the advantage of Momentum over SGD? (b) What is SoftMax activation? How is it different from the Sigmoid function? </td></tr>	3. (a)	Explain Adam optimizer algorithm using the function $f(w) = w^2$. Take the initial weight(w) is 2, learning rate 0.1, and exponential decay rates for the moments $\beta_1 = 0.9$ and $\beta_2 = 0.999$. What is the advantage of Momentum over SGD? (b) What is SoftMax activation? How is it different from the Sigmoid function?
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4.	(a)	Given a recurrent neural network for a sequential problem with k-dimensional inputs, n-dimensional hidden states, and d-dimensional outputs. If U is the weight matrix between input to hidden states, V is the weight matrix between hidden states to outputs and W is the weight matrix between hidden states then what is the dimension of U, V, and W? Mention the equations of hidden state s_t and output y_t .	3+3	CO3	L2
	(b)	Consider a GRU cell with Input at time step t is $\begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix}$ and previous hidden state is $\begin{bmatrix} 0.1 \\ 0.4 \end{bmatrix}$. Calculate the hidden state at time step t for this GRU cell. Assume all the weights are equal with $w = \begin{bmatrix} 0.6 & 0.2 \\ 0.1 & 0.7 \end{bmatrix}$ and bias $b = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$.			
5.	(a)	Define ResNet architecture. In ResNet-50, the bottleneck block uses 1×1 , 3×3 , and 1×1 convolution sequence. If the input to a block is 256 channels, and the middle 3×3 layer reduces it to 64 channels, what is the total number of parameters in this bottleneck block?	3+3	CO2	L3
	(b)	Discuss the architecture of VGG16. GoogLeNet uses an Inception module with 1×1 , 3×3 , and 5×5 convolutional layers. If an input feature map is $28 \times 28 \times 256$, and the number of filters in each convolutional branch is 64 (1×1), 128 (3×3), and 32 (5×5), what is the total number of output channels after concatenation?			

***** ALL THE BEST *****



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA-769008
MID-SEMESTER EXAMINATION, 2023

SESSION: 2023 – 2024 (Spring)
B. Tech. 8th / M. Tech. 2nd Semester

Subject code: CS6404
No. of pages: 2

Subject Name: Software Testing
Full Marks: 60

Dept. Code: CS
Duration: 2hrs

Figures at the right hand margin indicate marks.
All parts of a question should be answered at one place.
Answer all questions.

Q. No.	Particulars	Marks
1. (a)	A software was tested using error seeding strategy in which 40 errors were seeded in the code. When the code was tested using the complete test suite, 32 of the seeded errors were detected. The same test suite also detected 100 non-seeded errors. What is the estimated number of latent errors in the software after this testing is over?	2*10=20
(b)	Suppose, there are 3 input variables in a module. How many test cases can be designed using worst-case testing method?	
(c)	At least how many test cases are required to achieve MC/DC coverage of the following code segment if((a>10) OR (b<100)) AND ((c<50) OR (d>20)) then x = x+1;	
(d)	Give a positive and a negative test scenario for the “mobile number” field of the Student Registration Page in NIT Rourkela website.	
(e)	Suppose a program contains 5 decision points, each of which has 2 branches. How many test cases are necessary for branch testing?	
(f)	What is the relationship between cyclomatic complexity and program comprehensibility? Can you justify why such an apparent relationship exists?	
(g)	If there are 4 numbers of variables that have 3 values, then how many test cases are required to perform orthogonal array testing? Justify your answer.	
(h)	What do you mean by “ku” anomaly? What is its effect in data flow testing?	
(i)	If in a decision table, there are 3 rules and 4 binary conditions, then how many minimum and maximum test cases can be designed?	
(j)	What is JCUTE? Why it is used?	
2. (a)	Consider the following payroll system of an organization: (i) If the salary of an employee is less than or equal to Rs. 70,000 and expenses do not exceed Rs. 30,000, then 10% of the salary is deducted as income tax. (ii) If the salary is greater than Rs. 70,000 and less than or equal to Rs. 2,00,000 and expenses do not exceed Rs. 40,000, then 20% of the salary is deducted as income tax. (iii) For salary greater than Rs. 2,00,000, 5% additional surcharge is also charged, besides the income tax. (iv) If expenses are greater than Rs. 40,000, then 7% additional surcharge is also charged, besides the income tax. Design test cases for the above payroll system using cause-effect graph based technique.	12+8=20

(b) 3. (a) (b)	Consider a scenario in which we need to derive the test cases for a Banking Information System, that has four different functionalities as mentioned below: (i) Deposit money (ii) Withdraw money (iii) Transfer money (iv) Check balance These four functionalities can individually be <i>available</i> or <i>unavailable</i> or <i>partially available</i> to the users. Design test cases for the above Banking Information System using orthogonal array testing. Write a program in C for searching an integer value from a large stored sequence of integer values stored in an array of size 100, using linear search method. (i) Draw the CFG for the above program. (ii) Calculate the cyclomatic complexity of the program. (iii) List all the independent paths. (iv) Design all the test cases from the independent paths. Find out 4 possible mutants in the program developed according to Q 3(a). Check how many of them are killed by taking a set of arbitrary test data. Add new test data if required, to improve the mutation score.	12+8=20
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NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science and Engineering
Examination	Mid Semester – 2024-25
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	30
Duration	2 Hours

This question paper contains 2 pages

Instructions:

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**
1.	Apply black-box testing on a function named find-rectangle-intersection that accepts four floating point numbers representing two pairs of co-ordinates (x1, y1), (x2, y2), (x3, y3), (x4, y4) and generate the test cases. The first two points (x1, y1) and (x2, y2) represent the lower left and upper right points of the first rectangle. The second two points (x3, y3) and (x4, y4) represent the lower left and upper right points of the second rectangle. It is assumed that the length and width of the rectangle are parallel to either the x-axis or y-axis. The function computes the points of intersection of the two rectangles and prints their points of intersection.	5	CO-1	BL-3
2.	Consider a system having an FSM for a gaming software named Gumball with the following states and transitions: States: NoQuarter, HasQuarter, Sold, SoldOut, Error Transitions: NoQuarter to HasQuarter: addQuarter() NoQuarter to NoQuarter: turnCrank() HasQuarter to Sold: turnCrank() Sold to SoldOut: takeGumball() SoldOut to Error: turnCrank(), addQuarter() HasQuarter to Error: addQuarter() Sold to Sold: turnCrank() Sold to NoQuarter: takeGumball() Design test cases for the above FSM using state table based testing.	5	CO-1	BL-4
3.	Consider the operation of the following washing machine: <i>The washing machine waits for the “Start” switch to be pressed. After the user presses the “Start” switch the machine fills the wash tub with either hot or cold water depending upon the setting of the “Hotwash” switch. The water filling continues until the high level is sensed. The machine starts the agitation motor and continues agitating the wash tub until either the present timer expires or the user presses the “Stop” switch. After the agitation stops, the machine waits for the user to press the “Start Drying” switch. After the user presses the “Start Drying” switch the machine starts the hot air blower and continues blowing hot air into the drying chamber until either the user presses the “Stop” switch or the present timer expires.</i> Design test cases for the above washing machine using decision table based testing.	5	CO-1	BL-4

4.	Consider the following C function named <i>sort</i>. <pre> /* sort takes an integer array and stores it in ascending order */ void sort (int a[], int n){ int i,j; for(i=0;i<n-1;i++) for(j=i+1;j<n;j++) if(a[i]>a[j]) { temp=a[i]; a[i]=a[j]; a[j]=temp; } } </pre> Draw the CFG for the above function. Determine the cyclomatic complexity. List all independent paths. Design test cases to cover all independent paths.	5	CO-2	BL-3
5.	Consider the program given in Q 4. Generate four mutants and design test cases taking into account each mutant, using mutation testing. Calculate mutation score of your test suite. Can you improve further the obtained mutation score?	5	CO-2	BL-3
6.	Design a test suite using MC/DC testing for the following expression: $a \ \&\& \ (b \ \ c)$. Why MC/DC testing is preferred in software industries?	5	CO-2	BL-3

CO:** Course Outcome; *BL:** Bloom's Taxonomy Level.

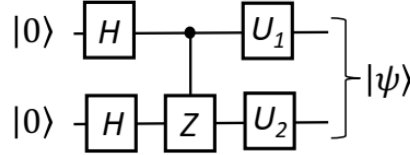
National Institute of Technology, Rourkela
Mid Semester Examination, Spring 2024
Quantum Computing (CS6520)

Duration: 2 Hours
Number of pages: 1

Full Marks: 30

- **Attempt all questions. Q1 – Q4.**
- **If you answer a question, all sub-questions should form part of it.**

- Q 1. (a) State and prove the no-cloning principle.
 (b) What should U_1 and U_2 be to prepare the entangled state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ in the following figure?

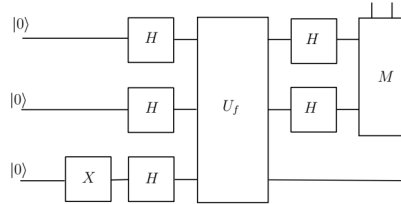


- (c) Show that $U = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ is unitary. (2 + 2 + 2)

- Q 2. In the quantum teleportation protocol, Alice and Bob are each in possession of one qubit of a pair in the joint state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. In addition, Alice has a qubit in an arbitrary state $|\phi\rangle$. Explain how the protocol works. In particular, show that it involves the transmission of exactly two classical bits of information from Alice to Bob and demonstrate how, at the end of the protocol, Bob is in possession of a qubit in state $|\phi\rangle$. (8)

- Q 3. (a) Write out the matrix for the three-qubit Toffoli gate.
 (b) A three-qubit system is in the state $\frac{1}{2}(|000\rangle + |101\rangle + |110\rangle + |111\rangle)$. The Toffoli gate is applied to this system and then a measurement is performed on the first qubit. What is the probability of observing a $|1\rangle$? Show all working out.
 (c) Suppose a two-qubit system is in the state $0.8|00\rangle + 0.6|11\rangle$. A Pauli X gate (i.e. a NOT gate) is applied to the second qubit, and a Hadamard gate is applied to the first qubit. What is the new state of the system? (1 + 5 + 2)

- Q 4. (a) Let $f : \{0, 1\}^2 \rightarrow \{0, 1\}$ be a Boolean function of two inputs. Let U_f be the implementation of f as a unitary operator on 3 qubits defined by: $U_f|x\rangle|y\rangle|z\rangle = |x\rangle|y\rangle|z \oplus f(x, y)\rangle$, where $\oplus$ denotes the exclusive-or operation, and $|x\rangle|y\rangle|z\rangle$ is any computational basis state. Consider the following circuit (a two-qubit version of the Deutsch-Josza circuit) in which X denotes a NOT gate, H denotes a Hadamard gate and M is a two-qubit measurement in the computational basis.



- (b) Show that if f is a constant function, the outcome of the measurement M is 00 with probability 1.
 (c) Show that if f is the XOR function, the outcome of the measurement M is 11 with probability 1.
 (d) What are the probabilities of M measuring 00 and 11 respectively, if f is the Boolean AND function?

(2 + 3 + 3)

*****All the best*****

National Institute of Technology, Rourkela
Mid Semester Examination, Spring 2025
Quantum Computing (CS6520/CS3012)

Duration: 2 Hours
Number of pages: 2

Full Marks: 30

- Attempt any **THREE** questions

Q 1. (a) Calculate

$$\begin{bmatrix} 3+2i & 5-i & 2i \\ 0 & 12 & 6-3i \\ 2 & 4+4i & 9+3i \end{bmatrix} \otimes \begin{bmatrix} 1 & 3+4i & 5-7i \\ 10+2i & 6 & 2+5i \\ 0 & 1 & 2+9i \end{bmatrix}.$$

[CO1, Apply]

- (b) Prove that the product of a doubly stochastic matrix with another doubly stochastic matrix is also a doubly stochastic matrix. [CO1, Analyze]
- (c) Consider the following hypothetical situation at a hypothetical college. Thirty percent of all math majors become computer science majors after one year. Another 60% become the physics majors after one year. After a year, 70% of the physics majors become math majors and 10% of the physics majors become computer science majors. In contrast to the other departments, computer science students are usually very happy: only 20% of them become math majors and 20% become physics majors after a year. (a) Draw a graph that describes the situation. (b) Give the corresponding adjacency matrix. (c) If a student is majoring in one of these three areas, indicate her probable major after 2 and 4 years. [CO1, Analyze]

(2 + 3 + 5)

Q 2. (a) Show that the matrix $\begin{bmatrix} 7 & 6+5i \\ 6-5i & -3 \end{bmatrix}$ is hermitian.

[CO1, Understand]

(b) Show that the matrix $\begin{bmatrix} \frac{1+i}{2} & \frac{i}{\sqrt{3}} & \frac{3+i}{2\sqrt{15}} \\ \frac{-1}{2} & \frac{i}{\sqrt{3}} & \frac{4+3i}{2\sqrt{15}} \\ \frac{1}{2} & \frac{-i}{\sqrt{3}} & \frac{3i}{2\sqrt{15}} \end{bmatrix}$ is a unitary matrix.

[CO1, Understand]

- (c) Let M be any n-by-n doubly stochastic matrix. Let X be an n-by-1 column vector. Let the result of $MX = Y$. Prove that if the sum of the entries of X is x, then the sum of the entries of Y is also x, i.e., M preserves the sum of the entries of a column vector multiplied at the right of M. [CO1, Analyze]
- (d) Show that if U is unitary, then so is C-U. [CO1, Analyze]

(2 + 2 + 4 + 2)

Q 3. (a) Do the vectors

$$A = [1+i, \quad 2-i]^T \quad \text{and} \quad B = [2+2i, \quad 1-2i]^T$$

represent the same state? Prove your answer.

[CO2, Analyze]

(b) What is $\sqrt{NOT}$ gate. Why is it called so??

[CO3, Understand]

(c) Assume that the particle is confined to $\{x_0, x_1, \dots, x_5\}$ and the current state vector is

[CO2, Apply]

$$|\psi\rangle = [2-i, \quad 2i, \quad 1-i, \quad 1, \quad -2i, \quad 2]^T$$

What is the likelihood of finding the particle at position x_3 ?

(d) Calculate the bra corresponding to the ket:

$$|\psi\rangle = [3+i, \quad -2i]^T$$

[CO1, Apply]

- (e) Find a matrix that corresponds to NOR.
- (f) Construct the NAND with one Toffoli gate.

[CO3,Apply]

[CO4,Apply]

(2 + 2 + 1 + 2 + 3)

Q 4. (a) Consider the following quantum operation:

$$(H \otimes I)U_f(H \otimes H)|0, 1\rangle.$$

- (i) Draw the quantum circuit corresponding to this operation.
- (ii) Determine the final state of the system before measurement.
- (b) Consider a qubit whose θ is equal to $\frac{\pi}{4}$. Change it to $\frac{\pi}{3}$ and picture the result. Then, compute its likelihood of collapsing to the south pole after being observed.

[CO4,Apply]

[CO4,Analyze]

[CO2,Apply]

(3 + 3 + 4)

*****All the best*****

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID SEMESTER EXAMINATION, 2024
MTech 2nd Semester and BTech 6th Semester

SUBJECT: Network Security
FULL MARKS: 30

DEPT. CODE: CS 6304
Duration of Examination: 2Hours
Number of pages: 1

Answer **Any five**. Figures at the right hand margin indicate marks
 All parts of a question should be answered at one place

Q.No.	Question	Marks
1	(a)Analyze Known-Message Attack and Chosen-Message Attack on RSA digital signature. (b) Describe how confidentiality is achieved in RSA digital signature.	[4+2]
2	(a) How the challenge-response authentication is achieved using keyed hash function? Explain with a suitable diagram. State merits and demerits of this approach. (b) How a claimant can verify itself to the verifier in entity authentication?	[4+2]
3	(a) Suppose, we apply the majority function on buffers A, B and C of SHA-512. If the leftmost hexadecimal digits of the three buffers are 2H, 1H and 0H respectively, what will be the leftmost digit of the result? (b) Explain why private-public keys can not be used in creating MAC. (c) In Random Oracle Model, why the oracle needs to make a note of the digest created for a message and gives the same digest for the same message?	[2+2+2]
4.	(a) Differentiate among password based, challenge-response and Zero Knowledge Authentication mechanisms Explain Fiat-Shamir protocol with a suitable diagram. (b) Why the idea of Merkel-Damgard scheme is very important for the design of cryptographic hash function?	[4+2]
5	(a) Explain Lamport's one time password mechanism with a suitable diagram. (b) What is the padding for SHA-512 if the length of the message is 6143 bits? Justify your answer.	[4+2]
6.	(a) Differentiate among HMAC and CMAC. How the CMAC is different from CBC mode of encipherment? (b) Assume the messages in a hash function are 9 bits long and it has a 6 bit digest. What is the possible number of messages? Show that at least one digest corresponds to $(k+1)$ messages, where k is a constant.	[4+2]

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID SEMESTER EXAMINATION, 2025
MTech 2nd Semester and BTech 7th Semester

SUBJECT: Network Security
FULL MARKS: 30

DEPT. CODE: CS 6304
Duration of Examination: 2Hours
Number of pages: 1

Answer **All questions**. Figures at the right hand margin indicate marks
All parts of a question should be answered at one place

Q. No.	Question	Marks	CO	BL
1	(a) Consider a system containing 1000 hosts. Each host runs 3 network applications (a file sharing application, voice call application, and instant messaging application), and each application should be able to communicate with the corresponding application on any other host (e.g. voice on host 1 with voice on host 2; but not voice on host 1 with file sharing on host 2). Answer the following questions with justifications. (i) If host-level symmetric key security is required in the network, what is the maximum number of session keys needed? (ii) If application-level symmetric key security is required in the network (for the 3 applications), what is the maximum number of session keys needed? (b) Can we use a checksum function as a cryptographic hash function? Justify your answer. Suppose, we apply the majority function on buffers A, B and C of SHA-512. If the leftmost hexadecimal digits of the three buffers are 4H, 2H and 1H respectively, what will be the leftmost digit of the result?	[3+3]	CO1	L2
2	(a) State two reasons of why the pre-distribution of symmetric keys is harder than that of public keys. What are the mandatory information available in an X.509 certificate? (b) Differentiate between data origin authentication and entity authentication. How a claimant can verify itself to the verifier in entity authentication?	[3+3]	CO1	L1
3	(a) State the major properties of public key ciphers. How authentication and confidentiality is achieved here? (b) How the Kerberos provides cryptographic protections against spoofing? Write the strengths of Kerberos.	[3+3]	CO2	L3
4.	(a) In station-to-station protocol, show that if the identity of the receiver is removed from the signature, the protocol becomes vulnerable to the man-in-the-middle attack. (b) Explain why private-public keys cannot be used in creating a MAC. Distinguish between an MDC and a MAC.	[3+3]	CO1	L2
5	Discuss the design objectives, algorithm and security of HMAC.	[6]	CO2	L4



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Department	Computer Science and Engineering
Examination	Mid Semester
Program	B.Tech/M.Tech//PhD
Session	Spring (2024-2025)
Course code & Course Name	Advanced Computer Networks (CS6302)
Marks	30
Duration	2 hours

This question paper contains 02 pages

Instructions:

1. **Attempt all the questions.**
2. **Write answers to all parts of a question in one place.**

Q. No.	Questions	Marks	CO	BL
1. (a)	In HTTP, draw a figure to show the application of cookies in a scenario in which the server uses cookies for advertisement. Use only three sites.	3	CO2	BL3
(b)	An HTML page is as follows: <html> <body> Click her for info </body></html> If the user clicks on the hyperlink, a TCP connection is opened and a series of lines is sent to the server. List all the lines sent.	2	CO2	BL2
(c)	Developers of Web browsers quickly discovered that sending all the objects in a Web page over a single TCP connection has a Head of Line (HOL) blocking problem. What is HOL blocking problem in HTTP/1.1? How does HTTP/2 attempt to solve it?	3	CO2	BL4
2. (a)	Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?	3	CO1	BL1
(b)	Consider a client and a server connected through one router. Assume the router can start transmitting an incoming packet after receiving its first h bytes instead of the whole packet. Suppose that the link rates are R byte/s and that the client transmits one packet with a size of L bytes to the server. What is the end-to-end delay? Assume the propagation, processing, and queuing delays are negligible. Generalize the previous result to a scenario where the client and the server are interconnected by N routers.	5	CO1	BL3

3. (a)	Suppose within your Web browser, you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of $RTT_1, \dots, RTT_n$. Further suppose that the Web page associated with the link contains exactly one object, consisting of a large amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. Assuming transmission duration of $0.002 * RTT_0$ of the object, how much time elapses from when the client clicks on the link until the client receives the object?	2	CO2	BL3
(b)	Under what circumstances is file downloading through P2P much faster than through a centralized client-server approach? Justify your answer.	3	CO1	BL2
4. (a)	What is meant by destination-based forwarding? How does this differ from generalized forwarding (which of the two approaches is adopted by Software-Defined Networking)?	2	CO3	BL1
(b)	Consider the SDN OpenFlow network shown in Figure 1. Suppose that the desired forwarding behavior for datagrams arriving at s_2 is as follows: - any datagrams arriving on input port 1 from hosts h_5 or h_6 that are destined to hosts h_1 or h_2 should be forwarded over output port 2; - any datagrams arriving on input port 2 from hosts h_1 or h_2 that are destined to hosts h_5 or h_6 should be forwarded over output port 1; - any arriving datagrams on input ports 1 or 2 and destined to hosts h_3 or h_4 should be delivered to the host specified; - hosts h_3 and h_4 should be able to send datagrams to each other. Specify the flow table entries in s_2 that implement this forwarding behavior. Figure 1	4	CO3	BL3
(c)	Describe the main role of the communication layer, the network-wide state-management layer, and the network-control application layer in an SDN controller.	3	CO3	BL2

CO: Course Outcome; **BL:** Bloom's Taxonomy Level.